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An Adhesive and Corrosion-Resistant Biomarker Sensing Film for Biosmart Wearable Consumer Electronics

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***Abstract*— The integration of electrochemical sensors in wearable consumer electronics enables monitoring the health status of individuals at molecular levels across the general population, and thus can play a critical role in transforming personalized and precision medicine. Previously, we devised a seamless integration strategy to interface disposable mediator-free enzymatic sensors—constructed on anisotropic conductive films (ACFs)—with consumer electronics. To illustrate the generalizability of our approach, here, we leverage ACF electrodes as a foundation to develop widely used mediator-based enzymatic sensors, which possess different underlying reaction mechanisms. Accordingly, we demonstrated the ACF-based sensor's anti-corrosive performance and its compatibility for integration with contact pads on both flexible and rigid substrates. To position this mediator-based sensor for untreated biofluid analysis, we adopted a post-calibration methodology to facilitate sensor surface conditioning. To demonstrate the clinical utility of our approach, a representative mediator-based enzymatic glucose sensor was developed and coupled with contact pads on a circuit board to analyze the changes in sweat glucose levels with respect to meal intake (n = 26). The generalizability of the ACF-based sensor development and integration strategy allows for its adoption to target a wide panel of biomarkers and to transform the wearable consumer electronics into biosmart platforms. [2020-0193]**

***Index Terms*— Biosmart consumer electronics, electrochemical sensing, Internet of Things, personalized and precision medicine, sweat metabolites, wearable sweat analysis.**

I. Introduction

THE exponential proliferation of the Internet-of-Things devices is expected to lead to more than 1,000 sensors per person over the next decade [1]. Adaptation of a subset of such sensing devices toward health monitoring applications can create an unprecedented opportunity: realizing personalized and precision medicine through the large-scale and longitudinal collection of physiologically-relevant multi-dimensional datasets. Unlike the physical sensors that have been monolithically-embedded within consumer electronics, the electrochemical sensors are mostly implemented as disposable units, which are connected to corresponding readout modules (implemented as printed circuit boards, or PCBs) via rigid connectors [2]. Accordingly, the integration challenges, arising from the contradictory requirements for the disposability of the sensors and their monolithic implementations, have posed a major bottleneck to the incorporation of electrochemical sensors in wearable consumer electronics. In that regard, we previously devised an integration strategy, which transforms the contact pads on PCBs (the core of wearable consumer electronics) into interconnection substrates that directly interface with disposable electrochemical sensors (eliminating the need for external connectors) [3]. This integration strategy was demonstrated in the context of mediator-free enzymatic sensors on ACF-based substrates. To illustrate the generalizability of our approach, we developed mediator-based (Prussian blue-based) enzymatic glucose and lactate sensing interfaces on ACF electrodes. Within this context, we demonstrated immunity of our sensor-electronics integration against corrosion (a common challenge when interfacing electronics with biofluids [4]). Furthermore, we illustrated our sensor's integration compatibility with contact pads on both flexible and rigid substrates. To position this mediator-based sensor for untreated biofluid analysis, we adopted a post-calibration methodology to facilitate sensor surface conditioning, thus overcoming the susceptibility of the Prussian blue-based sensor response to interfering ionic species [5].

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II. Results and Discussion

A. Characterization and Integration of the Anisotropic Conductive Sensing Film

In our approach, we leverage a double-sided adhesive ACF (~50 μm-thick). One side of the adhesive film serves as the integration interface, where it can be directly adhered to PCB contact pads, realizing a μm-level vertical sensor-circuit signal interconnection. On the other side of the ACF, the biosensing interface can be constructed, by first patterning a noble metal layer, followed by modifying the recognition layers. The vertical conductivity of the ACF is rendered by metal particles (e.g., silver) that are embedded in the polymer matrix of ACF, as seen from the scanning electron microscopy (SEM) image in Fig. 1(a). By leveraging the adhesive coating of the ACF and without the need for a metal adhesion layer, a gold layer (Au) can be directly evaporated on the film's surface (200 nm-thick) as shown in Fig. 1(b), serving as a base for the subsequent chemical modification layers required to achieve biosensing. The adhesive coating on the other side of the ACF facilitates the adhesion of the ACF to contact pads, thereby forming the mechanically robust and electrically-conductive sensor-circuit interconnection.

Standard metal contact pads on a PCB galvanically corrode in an ionic biofluid solution during electrochemical sensing operation. This corrosion manifests as signal degradation and/or large background current (Fig. 1(c), corrosion current indicated in red). As shown in

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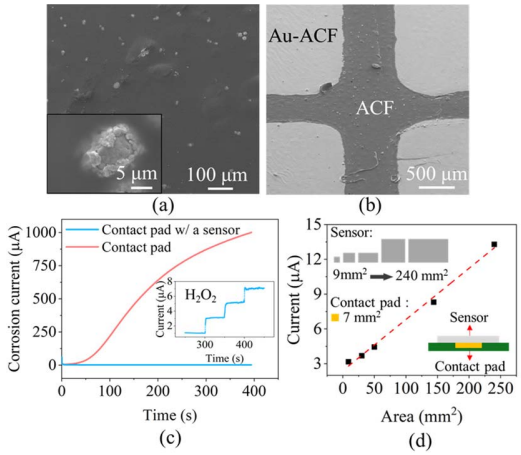


Fig. 1. (a) SEM image of the ACF. The inset shows a silver particle (conductive agent) embedded in the ACF matrix. (b) SEM image of the Au-patterned ACF. A liner-based shadow mask was used in conjunction with E-beam evaporation to form the desired pattern. (c) Amperometric characterization performed with PCB contact pads (immersion gold) with and without the ACF-based sensing unit in phosphate buffered saline (PBS, 0.5 V vs. Ag/AgCl). The inset shows the electrochemical response of the ACF-coupled PCB interface to sequential H_2O_2 additions (3 steps, 100 μM per step) PBS solution (Ag/AgCl as the reference electrode). (d) Current response of ACF-based sensing units with various surface areas, measured via the same PCB contact pad.

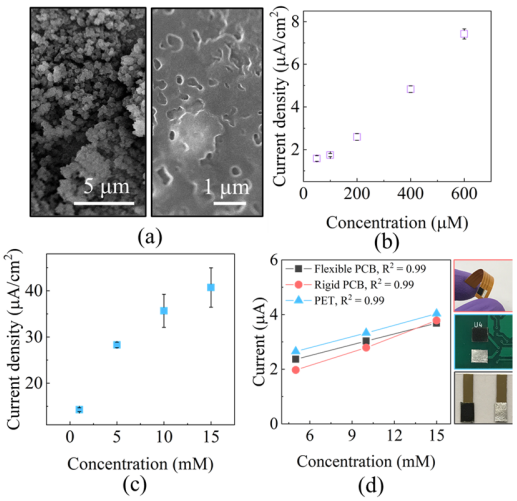


Fig. 2. (a) SEM image of the deposited Prussian blue layer (left) and glucose oxidase-chitosan layer (right). (b-c) Calibration curves of ACF/PCB-based (b) glucose sensors and (c) lactate sensors (SE, $n = 3$, performed in PBS under stirring condition). (d) Responses of lactate sensors taped onto contact pads of a flexible PCB, rigid PCB and PET substrates (ACF size: 3 mm $\times$ 4 mm).

Fig. 1(c), the PCB contacts (based on immersion gold, as a cost effective solution) coupled with an ACF-based metal film (platinum/Au) presented near-zero corrosion current (blue line), indicating that the galvanic corrosion underneath the metal contact pads was effectively prevented. The amperometric characterization of the ACF/PCB-based electrode response to a series of oxidant solutions (here, H_2O_2) indicated the preservation of the catalytic capability of the electrode (Fig. 1(c), inset). The excellent anti-corrosion performance of the ACF sensing film can be attributed to: (1) the hydrophobicity of the adhesive coating on both sides of the film, and (2) the absence of oxidation prone-adhesion metal (e.g., chromium). Additionally, as shown in Fig. 1(d), amperometric characterization using ACF films with different areas on a fixed PCB contact pad yielded a linear relationship between the corresponding electrochemical current responses (toward H_2O_2) and the ACF sensing area. This observation illustrates the high degree of freedom achieved by our approach for tuning the sensor's output current level, simply by customizing the area of the ACF sensing film and (without adjusting the circuitry or contact pads designs).

B. Electrochemical Sensor Fabrication

To demonstrate the generalizability and suitability of the ACF as a base to form biosensing interfaces, we realized ACF-based glucose and lactate sensors, following a commonly-used wearable enzymatic sensor development protocol [6]. Accordingly, first, Prussian blue was electrodeposited on an 12 mm^2 Au-patterned ACF electrode (Fig. 2(a)) to improve the selectivity of the sensor by decreasing the required applied potential for the electroanalysis of the end product of oxidase-based reactions (here, H_2O_2) [2]. Then, glucose/lactate oxidase, embedded within a chitosan matrix, was coated on the Prussian blue-deposited ACF to facilitate enzymatic reactions in the presence of glucose/lactate molecules (Fig. 2(b) and (c)). The sensitivity levels of our ACF glucose and lactate sensors (coupled with PCB contact pads) were respectively 10.8 and 1.8 $\mu A{\cdot}mM^{-1}{\cdot}cm^{-2}$, which are comparable with the performance of previously reported Prussian blue-based sensors constructed on conventional substrates [7], [8]. To present the versatility of our adhesive sensing approach, the developed lactate sensor was adhered onto the contact pads of a rigid PCB, flexible PCB, and polyethylene terephthalate (PET), and for each case the electrochemical response was characterized. Fig. 2(d) illustrates that the integrated sensors' responses to increasing lactate concentrations were proportional and on the same levels in all three cases (despite different underlying contact pad materials and substrates).

C. Application of Glucose Sensors for Sweat Analysis

Figure 3(a) demonstrates the accuracy of the devised glucose sensors, where we performed a blind study by analyzing a series of glucose-spiked PBS samples in random sequence by ACF-based sensing films (taped onto PCB contact pads) and a gold-standard laboratory instrument (YSI 2900D). To adapt this enzymatic sensor for sweat analysis, the susceptibility of Prussian blue-based sensor response to interfering species must be addressed [5]. Here, we particularly performed calibrations after sensor exposure to sweat, which effectively results in conditioning the surface by the adsorption of the species present in the complex biomatrix. As shown in Fig. 3(b), this approach renders closely matched calibration curves (sensitivity: 13.3 $\mu A{\cdot}mM^{-1}{\cdot}cm^{-2}$) and accurate estimated target concentration in sweat (evident from the observed consistent sweat readings). To illustrate the clinical utility of the developed sensor, 26 sweat samples were obtained from different subjects before and after meal intake (following our reported sweat glucose study protocol [6]). Each sample was measured ex-situ three times by conditioned sensors. The results from the ex-situ analysis indicate that sweat glucose levels were elevated upon food intake for all subjects ($p < 0.001$, two-tailed paired t test, Fig. 3(c)).

III. Materials and Methods

A. Materials and Reagents

All reagents were purchased from Sigma-Aldrich (MO, USA) unless stated. Phosphate buffered saline (PBS, 1×, Gibco, pH 7.2) and isopropyl alcohol were purchased from Fisher Scientific

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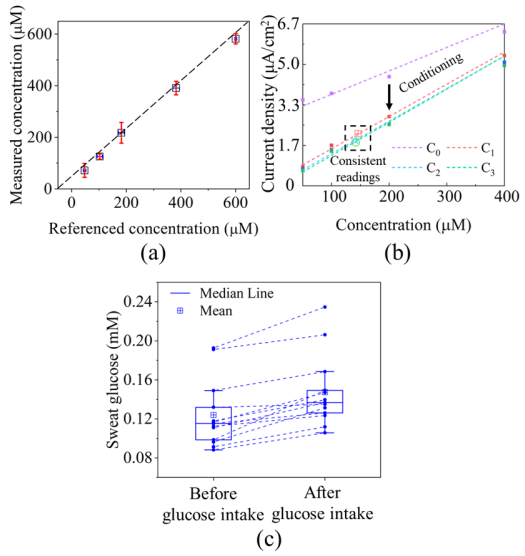


Fig. 3. (a) Estimated glucose concentration of a random sequence of glucose-spiked PBS samples by the ACF/PCB-based glucose sensor vs. a gold-standard lab instrument (SE, n = 3). (b) Calibration curves of a glucose sensor before (C_0) and after 1st (C_1), 2nd (C_2), and 3rd (C_3) exposure to sweat. The three estimated glucose levels of a sweat sample are marked with a dotted line box (indistinguishable, because they overlay). (c) Sweat glucose concentration before and after meal intake ($n = 26$), measured by the ACF/PCB-based glucose sensor.

(MA, USA). Anisotropic conductive film (9703, 50 μm, ACF) was purchased from 3M (MN, USA).

B. Electrode Preparation and Electrochemical Tests

Experimental procedures regarding the Au- and Pt-metallization of the ACF were detailed in our previous work [3]. To fabricate Prussian blue-based glucose/lactate sensors, Prussian blue was electrochemically deposited onto the Au-ACF electrode by performing cyclic voltammetry from 0 to 0.5 V (vs. Ag/AgCl) for 10 cycles at 20 mV s^{-1} in a fresh Prussian blue solution and dried on a hot plate at 100 °C for 1 h. To develop the glucose/lactate sensors, the glucose/lactate oxidase was immobilized onto the PB-based electrode surface by utilizing chitosan [5]. Constant potential amperometric measurements were conducted at −0.1 V (vs. Ag/AgCl). The sensor characterization and PBS/sweat sample analysis were performed by adhering the sensors on the contact pads of a PCB, which in turn was connected to a potentiostat (CHI 660E).

C. Sweat Stimulation and Collection

Sweat secretion stimulation was performed (Macroduct®Sweat Collection System) in subjects after 12 h fasting and immediately after glucose intake (30 g). For each case, the secreted sweat samples were collected for 30 mins and stored in −20 °C freezer before ex-situ analysis by our sensors.

IV. Conclusion

We demonstrated an adhesive and corrosion-resistant biomarker sensing film, which can transform contact pads on flexible and rigid substrates (PET, FPCB, and PCB) into electrochemical sensing interfaces, thus enabling seamless integration of disposable electrochemical sensors with wearable consumer electronics. Amperometric characterization illustrated the immunity of the integrated sensing interface to corrosion in high salt buffer solutions. The generalizability and utility of the biosensing functionality was demonstrated by developing widely used mediator-based (Prussian blue) glucose and lactate enzymatic sensors. This demonstration particularly informs the potential utility of our approach in emerging applications that rely on self-powered biosensors and biofuel cells (both of which leverage the unique properties of Prussian blue and enzymatic reactions). To position our mediator-based sensor for untreated biofluid analysis, we adopted a post-calibration methodology to facilitate sensor surface conditioning, thus overcoming the susceptibility of the Prussian blue-based sensor response to interfering ionic species. The devised ACF-based sensors can be tailored to target a wide panel of biomarkers by using different recognition layers (e.g., different oxidase-based enzymes). The range of bioanalytical capabilities achieved by our sensors and their ease of integration pave the path for the transformation of wearable consumer electronics into biosmart platforms.

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